

HAT-P-36b

R-0756

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260215 · created 2026-07-28T03:31:58+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461086.9541 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.011
Transit depth (Rp/Rs)^2	2.18 +/- 0.32 [%]
Orbital Inclination (inc)	83.93 +/- 2.35
Airmass coefficient 1 (a1)	0.981 +/- 0.013
Airmass coefficient 2 (a2)	0.016 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#1 - [92, 42]
Optimal Aperture	3.48
Optimal Annulus	11.21
Transit Duration (day)	0.0898 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.011 vs 0.126 (deviation 1.98 σ)
Duration fit vs published	0.0898 d vs 0.0925 d (deviation 0.33 σ)
Mid-time O-C	13.5 min
Depth SNR	13.3
Residual scatter	1.3 %

Light curve

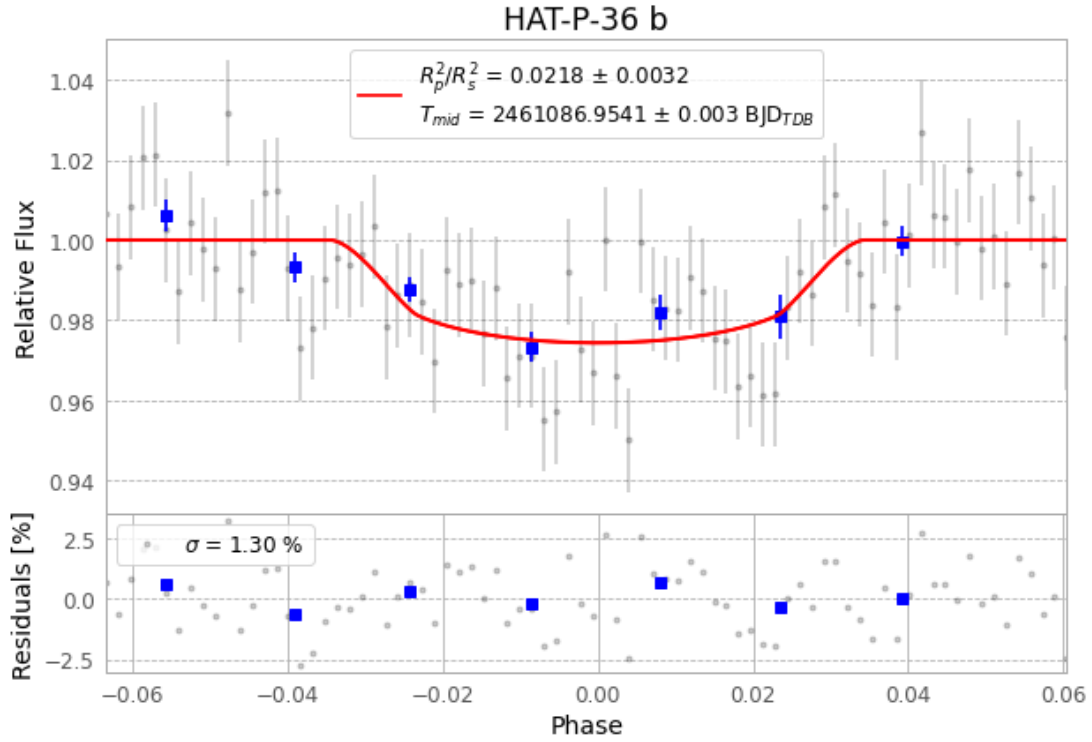


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-02-15.png (SHA-256 8b6cb43c5562937a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 108], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2557eaa0748f65f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), HATP-36260215084516.FITS → HATP-36260215124215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260215.log (528c01ac3291...), FinalLightCurve_HAT-P-36 b_2026-02-15.png (8b6cb43c5562...), FinalLightCurve_HAT-P-36 b_2026-02-15.csv (b632cd7e3cdf...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 03:31Z.